

Classwork (23rd June)

1. Suppose a researcher wants to (1) determine the percentage of neurodivergent people in the Dhaka Division and (2) investigate whether neurodivergent people with a high hip-to-waist ratio are more likely to be obese in this area. To address these two research questions, he collected data from 125 people from each of the eight thanas in the eastern part of the Dhaka Division. From each person included in the sample, he collected data on neurodivergence (yes/no), BMI (kg/m^2), gender (male/female), age (in years), type of residence (urban/rural), socioeconomic status (poorest, poor, middle, rich, richest), diet (vegan, vegetarian, others), and daily food consumption (kcal).

It was found that 17% of people are neurodivergent in the mentioned area, and there is a significant association between hip-to-waist ratio and obesity.

- Are study questions (1) and (2) descriptive or inferential?
- Define the population, sample and the total sample size for the study questions.
- What are the parameter and the statistic for study question (1)?
- Did the researcher work with primary data or secondary data?
- Classify the variables as qualitative/quantitative, discrete/continuous, and specify the scale of measurement for each variable.
- The BMI (kg/m^2) values of the sampled individuals are summarised in the following grouped frequency distribution:

BMI (Kg/m^2)	Frequency
15–20	50
20–25	310
25–30	360
30–35	210
35–40	70

- Calculate the mean, median and mode for the above grouped data. Interpret your findings.
- Draw a histogram and a frequency polygon on the same graph for the BMI data.
- Draw an ogive and find the percentage of people who have a BMI less than or equal to 28 kg/m^2 .
- If the mean BMI for neurodivergent people is 22.5 kg/m^2 , then find the mean BMI for normal people (hint: combined mean).

- g) The socioeconomic status (SES) of 25 neurodivergent female is recorded below:

Poorest	Middle	Rich	Rich	Poorest
Richest	Poorest	Rich	Middle	Middle
Poorest	Poorest	Middle	Middle	Poor
Richest	Richest	Poor	Poorest	Poor
Rich	Richest	Poor	Richest	Rich
Middle	Poor	Richest	Rich	Poorest

- (i) Construct a frequency distribution table for the socioeconomic status of these individuals.
 - (ii) Calculate the percentage of individuals who belong to the *Richest* category.
 - (iii) Draw an appropriate graph to present the data. Discuss your findings.
- h) The daily food consumption (in kcal) of the top nine individuals in terms of food intake is given below:

2800, 2950, 3100, 3250, 3400, 3550, 3700, 3900, 8500

- (i) Which measure of central tendency would be most appropriate to summarise the typical daily food consumption in this case? Justify your answer. Find that measure and interpret the result.